

Drone Soccer: Learning to Manipulate with Multicopter Downwash

Neelay Joglekar¹, Bavin Saravanan¹, Yutong Wang¹, Varun Kandiyappan¹, Junyi Geng², and Sebastian Scherer¹

¹Robotics Institute, Carnegie Mellon University, Pittsburgh USA

²Aerospace Engineering Department, Penn State University, University Park USA

Introduction

- Unmanned Aerial Manipulators (UAMs) utilize direct contact (e.g. arms, cables) for manipulation.
- Is direct physical contact the only interaction mode available for UAMs?



- Drones constantly interact with their surroundings via “downwash,” the airflow field produced by drone’s propellers.
- We design and demonstrate a **drone soccer task**, where the drone must actively utilize downwash to perform manipulation.

Method

We develop a **downwash dynamics model** to train an RL policy to **push a ball towards a goal**.

$$U_H = \sqrt{\frac{1 \cdot u}{2\rho A_p N}}$$

N : # props $u \in \mathbb{R}^N$: Prop torques
 ρ : Air density A_p : Prop disk area
 $U_H \in \mathbb{R}$: Induced speed

1. Induced Speed Estimation

Airflow speed at rotor plane.

2. Downwash Speed Estimation

Computed from Boursfeld et al’s model [1].

3. Ground Spreading Heuristic

Airflow radiates outwards near ground.

$$\phi = \arctan \frac{r}{e_3 \cdot \mathbf{T} \begin{bmatrix} r \cos \theta & r \sin \theta & s \end{bmatrix}^\top}$$

$$\vec{v} = \begin{bmatrix} \sin \phi \cos \theta & \sin \phi \sin \theta & -\cos \phi \end{bmatrix}^\top$$

$\mathbf{T} \in SE(3)$: Local-to-world transform $e_3 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^\top$
 ϕ : Spreading angle $\vec{v} \in \mathbb{R}^3$: Downwash velocity direction

4. Ball Force Estimation

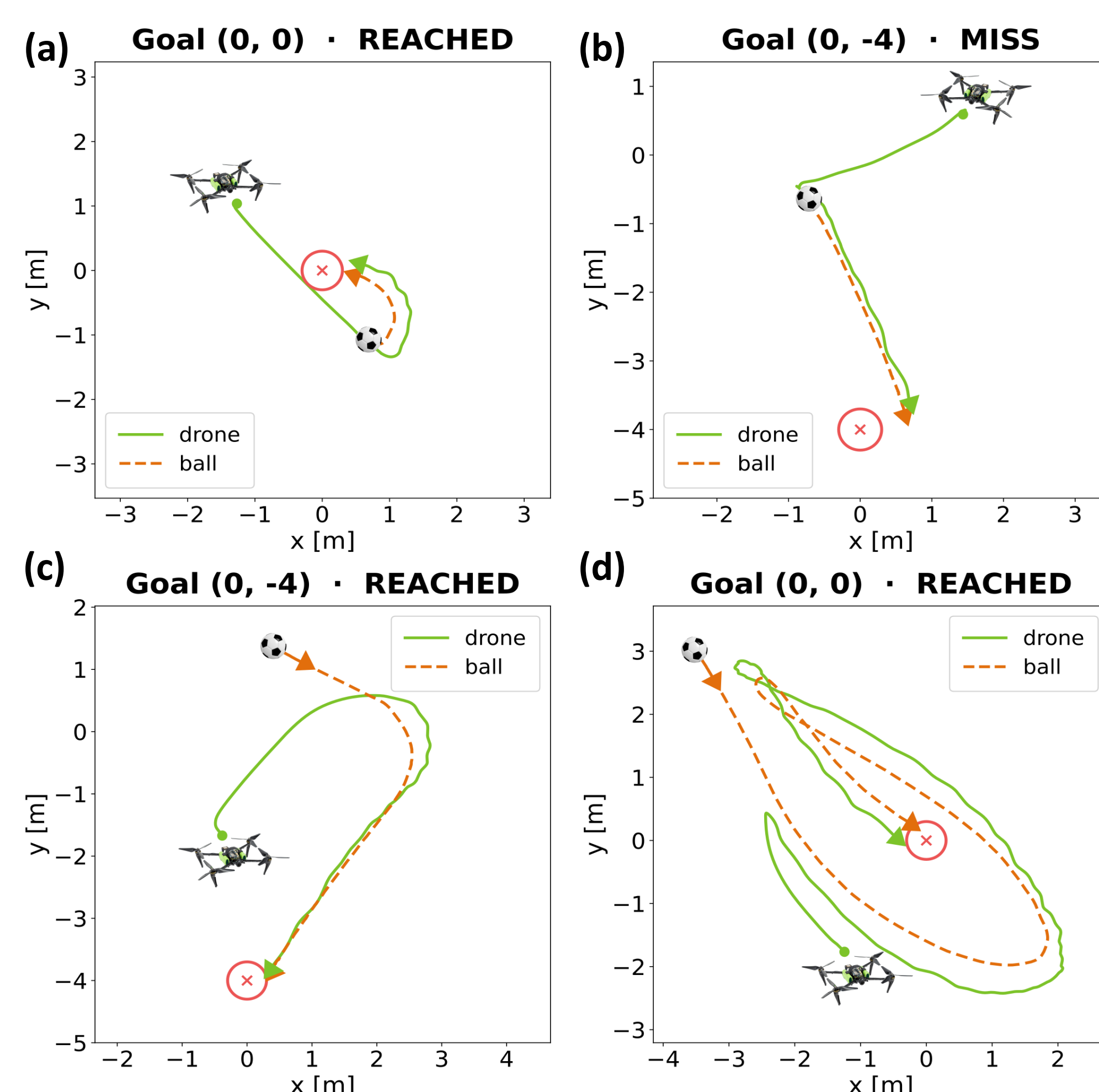
Blunt drag model.

5. Reinforcement Learning Setup

- State space**: drone inertial, ball inertial, goal location.
- Action space**: 10 Hz position setpoints.
- Reward terms**: dense ball progress rewards, sparse goal reward, sparse crash penalty, dense setpoint jitter penalty.
- Optimizer**: PPO.
- Curriculum**: initial ball velocity and scene configuration increasingly randomized throughout training.



Results

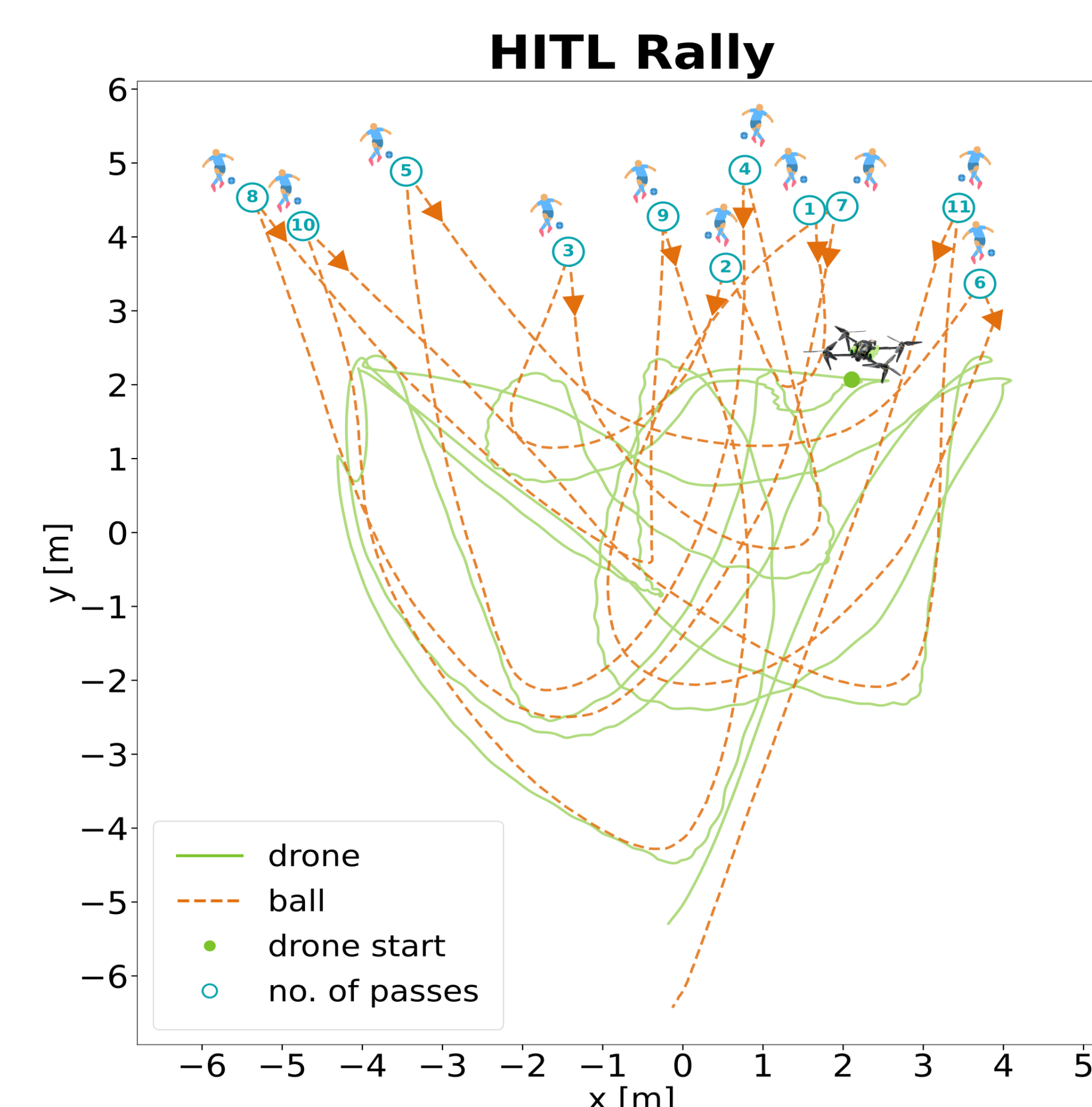


- We directly deploy our policy on hardware
- Policy reaches goal in **3 out of 4** scene configurations.
- Human-in-the-loop (HITL) experiment**: The drone and human subject achieve a **10-pass rally**.
- Policy occasionally struggled to redirect ball, likely due to sim-to-real gap.

	(a)	(b)	(c)	(d)
Goal Reached?	True	False	True	True
Median Ball Speed (m/s)	0.43	0.50	0.90	0.91
Initial Ball Speed (m/s)	0.0	0.0	0.98	1.63
Final Ball Speed (m/s)	0.72	1.41	1.60	1.20

HITL Average Ball Speed Statistics (m/s)

	Min	Median	Max Successful	Max
	0.62	1.22	1.52	2.52



Discussion

- First policy to actively use downwash for manipulation.
- Next steps and Extensions:
 - On-robot reinforcement learning via self-play.
 - Multi-agent soccer and adversarial games.
 - Extend to leaf blowing, litter collection, animal herding.

References

[1] Boursfeld, Leonard, et al. "Robotics meets fluid dynamics: A characterization of the induced airflow below a quadrotor as a turbulent jet." IEEE Robotics and Automation Letters 10.2 (2024): 1241-1248.

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